

Assignment-2

1. Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.

Data Members(properties): Name Roll No Degree Hostel Current CGPA

Member Function(behavior): addDetails(); updateDetails(); updateCGPA(); updateHostel(); displaydetails();

Code:-

```
ques1.cpp
1 //Assignment- 2
2 //Question -1
3 #include <iostream>
4 using namespace std;
5 struct Student {
6     string name, degree, hostel;
7     int rollNo;
8     float cgpa;
9     void addDetails() {
10         cout << "Enter Name, Roll, Degree, Hostel, CGPA:\n";
11         cin >> name >> rollNo >> degree >> hostel >> cgpa;
12     }
13     void updateDetails() {
14         cout << "Update Name: ";
15         cin >> name;
16     }
17     void updateCGPA() {
18         cout << "Enter new CGPA: ";
19         cin >> cgpa;
20     }
21     void updateHostel() {
22         cout << "Enter new Hostel: ";
23         cin >> hostel;
24     }
25     void displayDetails() {
26         cout << name << " " << rollNo << " "
27             << degree << " " << hostel
28             << " CGPA: " << cgpa << endl;
29     }
30 };
31 int main() {
32     Student s;
33     s.addDetails();
34     s.displayDetails();
35     s.updateCGPA();
36     s.displayDetails();
37 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  X + v
Enter Name, Roll, Degree, Hostel, CGPA:
Raju 101 BTech A 9.8
Raju 101 BTech A CGPA: 9.8
Enter new CGPA: 8.5
Raju 101 BTech A CGPA: 8.5

-----
Process exited after 28.95 seconds with return value 0
Press any key to continue . . .
```

2. Differentiate between private and public access/scope. Perform the question no. 1 with class instead of structure with having the data members private and some member functions in private scope and some in public scope.

Code:-

```
ques1.cpp  ques2.cpp
1  //Assignment- 2
2  //Question -2
3  #include <iostream>
4  using namespace std;
5  class Student {
6  private:
7      string name, degree, hostel;
8      int rollNo;
9      float cgpa;
10 void secretFunction() {
11     cout << "Private function accessed\n";
12 }
13 public:
14 void addDetails() {
15     cin >> name >> rollNo >> degree >> hostel >> cgpa;
16 }
17 void updateCGPA() {
18     cin >> cgpa;
19 }
20 void displayDetails() {
21     secretFunction(); // private called inside public
22     cout << name << " " << rollNo << " "
23         << degree << " " << hostel
24         << " CGPA: " << cgpa << endl;
25 }
26 };
27 int main() {
28     Student s;
29     s.addDetails();
30     s.displayDetails();
31 }
```

Output:-

```
C:\Users\asus\Desktop\oops\ × + v
Raj 101 BTech A 9.8
Private function accessed
Raj 101 BTech A CGPA: 9.8

-----
Process exited after 45.01 seconds with return value 0
Press any key to continue . . .
```

3. Create a code snippet that illustrates the following: Calling of private member functions inside public member function

Code:-

```
ques1.cpp ques2.cpp ques3.cpp
1 //Assignment- 2
2 //Question -3
3 #include <iostream>
4 using namespace std;
5 class Demo {
6 private:
7     void secret() {
8         cout << "Private function\n";
9     }
10 public:
11     void show() {
12         secret(); // allowed
13     }
14 };
15 int main() {
16     Demo d;
17     d.show();
18 }
```

Output:-

```
C:\Users\asus\Desktop\oops\ × + v
Private function

-----
Process exited after 0.2419 seconds with return value 0
Press any key to continue . . .
```

4. Define a class Rectangle with variables width and height of integer typwe along with following:

- (a) void getdata() to initialize object values
- (b) void calculatearea() to calculate and display the area.

Code:-

```
ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp
1  //Assignment- 2
2  //Question -4
3  #include <iostream>
4  using namespace std;
5  class Rectangle {
6      int width, height;
7  public:
8      void getdata() {
9          cin >> width >> height;
10     }
11     void calculatearea() {
12         cout << "Area: " << width * height;
13     }
14 };
15 int main() {
16     Rectangle r;
17     r.getdata();
18     r.calculatearea();
19 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  x  +  v
10
20
Area: 200
-----
Process exited after 7.817 seconds with return value 0
Press any key to continue . . .
```

5. Define a class Complex with variables real and imaginary along with following:

- (a) void setComplex (float, float) to initialize object values.
- (b) void displayComplex() to show the complex number
- (c) Pass and return objects to calculate sum of two complex numbers. Display the sum.

Code:-

```
ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp  ques5.cpp
1  //Assignment- 2
2  //Question -5
3  #include <iostream>
4  using namespace std;
5  class Complex {
6      float real, imag;
7  public:
8      void setComplex(float r, float i) {
9          real = r;
10         imag = i;
11     }
12     void displayComplex() {
13         cout << real << " + " << imag << "i\n";
14     }
15     Complex add(Complex c) {
16         Complex temp;
17         temp.real = real + c.real;
18         temp.imag = imag + c.imag;
19         return temp;
20     }
21 };
22 int main() {
23     Complex c1, c2, c3;
24     c1.setComplex(2,3);
25     c2.setComplex(4,5);
26     c3 = c1.add(c2);
27     c3.displayComplex();
28 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  ×  +  v
6 + 8i
-----
Process exited after 0.2818 seconds with return value 0
Press any key to continue . . .
```

6. Implement scope resolution operator :: for the following uses:

- (a) Class functions defined outside the class
- (b) Access a global variable with same name as a local variable
- (c) Access a static variables
- (d) Use inbuilt libraries (cin and cout with scope resolution operator)

Code:-

ques1.cpp ques2.cpp ques3.cpp ques4.cpp ques5.cpp ques6.cpp

```
1 //Assignment- 2
2 //Question -6
3 #include <iostream>
4 using namespace std;
5 int x = 100; // global
6 class Demo {
7     static int count;
8 public:
9     void show();
10 };
11 int Demo::count = 0; // static
12 void Demo::show() { // outside class
13     int x = 10;
14     cout << "Local x: " << x << endl;
15     cout << "Global x: " << ::x << endl;
16     cout << "Static count: " << count << endl;
17 }
18 int main() {
19     Demo d;
20     d.show();
21     std::cout << "Using std::cout\n";
22 }
```

Output:-

```
C:\Users\asus\Desktop\oops\ >
Local x: 10
Global x: 100
Static count: 0
Using std::cout

-----
Process exited after 0.2797 seconds with return value 0
Press any key to continue . . .
```

7. Create a code to implement the *namespace* and use similar variables and functions defined in different code sections.

Code:-

```
ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp  ques5.cpp  ques6.cpp  [*] ques7.cpp
1  //Assignment- 2
2  //Question -7
3  #include <iostream>
4  using namespace std;
5  namespace A {
6      int x = 10;
7      void show() {
8          cout << "Namespace A: " << x << endl;
9      }
10 }
11 namespace B {
12     int x = 20;
13     void show() {
14         cout << "Namespace B: " << x << endl;
15     }
16 }
17 int main() {
18     A::show();
19     B::show();
20
21     cout << "A x: " << A::x << endl;
22     cout << "B x: " << B::x << endl;
23 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  X  +  v
Namespace A: 10
Namespace B: 20
A x: 10
B x: 20

-----
Process exited after 0.2887 seconds with return value 0
Press any key to continue . . .
```